CO315 - Computer Networks

Module-1

1-1 DATA COMMUNICATION

The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communication is the exchange of data between two devices via some form of transmission medium such as a wire cable.

Data Communication

The effectiveness of a data communications system depends on four fundamental characteristics.

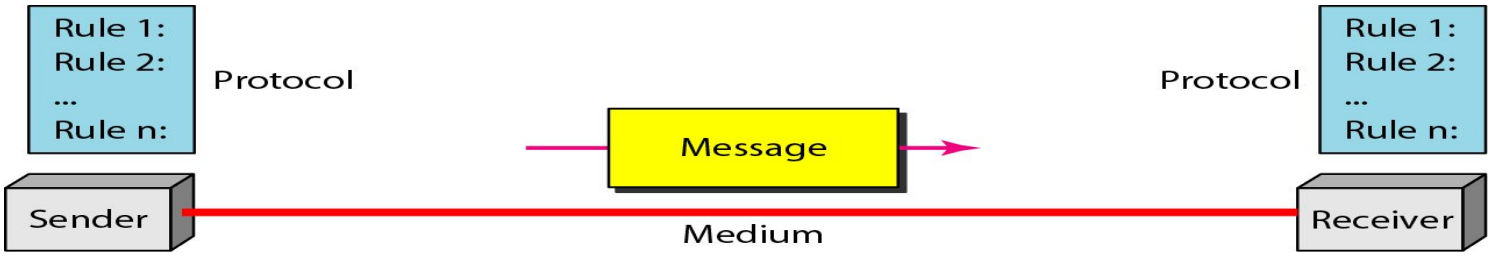
1. Delivery. The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

2 Accuracy. The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

3. Timeliness. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called *real-time* transmission.

4. Jitter. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Components of a data communication system



Message. The message is the information (data) to be communicated.

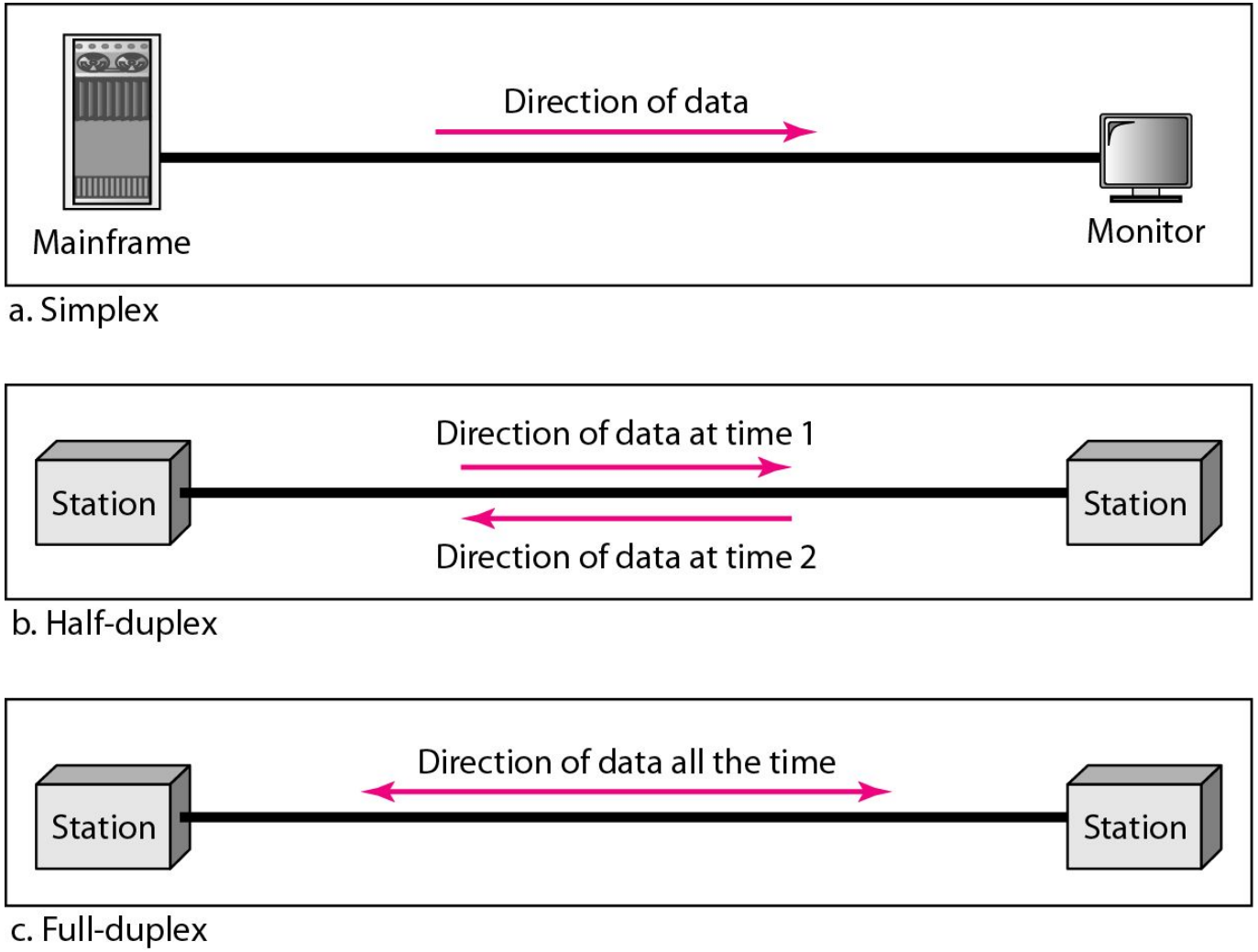
Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

Receiver. The receiver is the device that receives the message.

Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Ex- twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



Simplex: In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. Ex. Keyboard, Monitor.

Half-Duplex:

In half-duplex mode, each station can both transmit and receive, but not at the same time. :

When one device is sending, the other can only receive, and vice versa.

Ex. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

Full-Duplex:

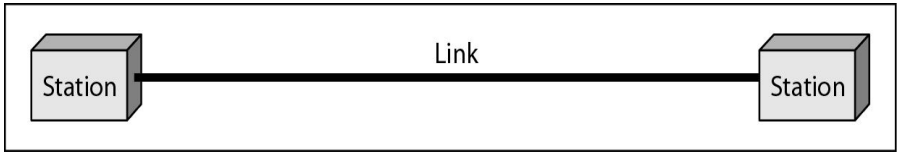
In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

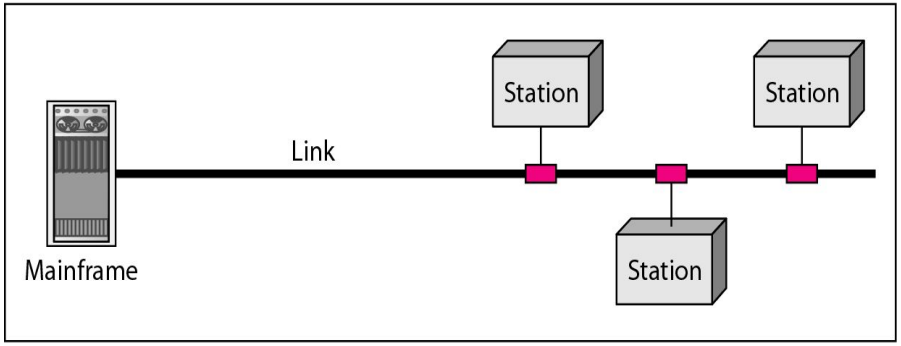
NETWORK

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network. A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

Types of connections: point-to-point and multipoint



a. Point-to-point



b. Multipoint

Point-to-Point A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices.

Multipoint A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.

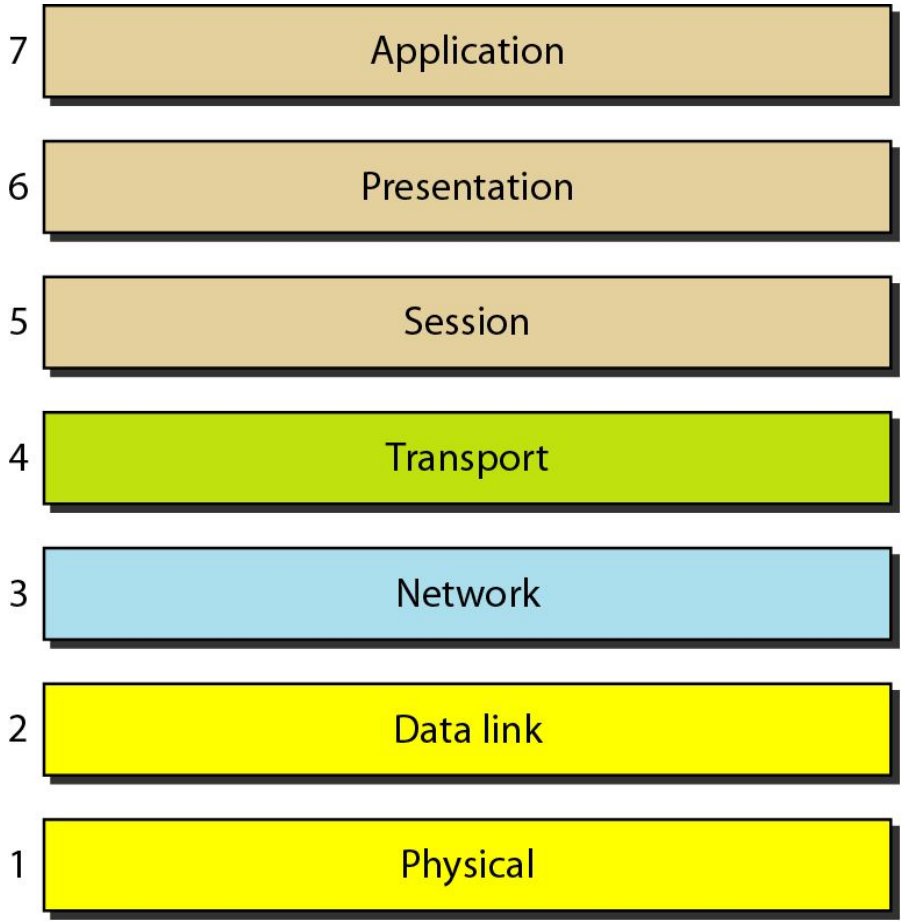
THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

Note

ISO is the organization.
OSI is the model.

Seven layers of the OSI model



The interaction between layers in the OSI model

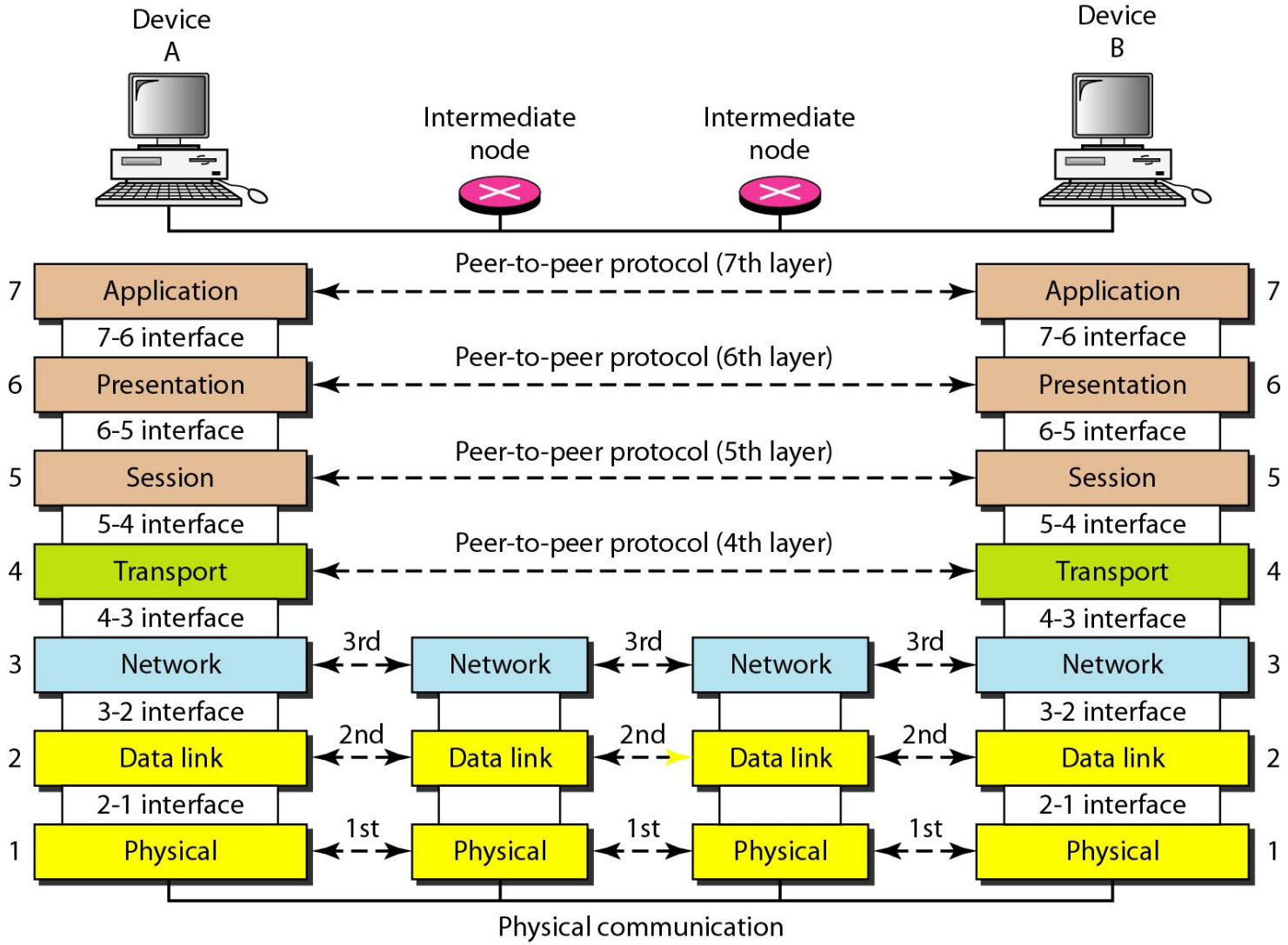
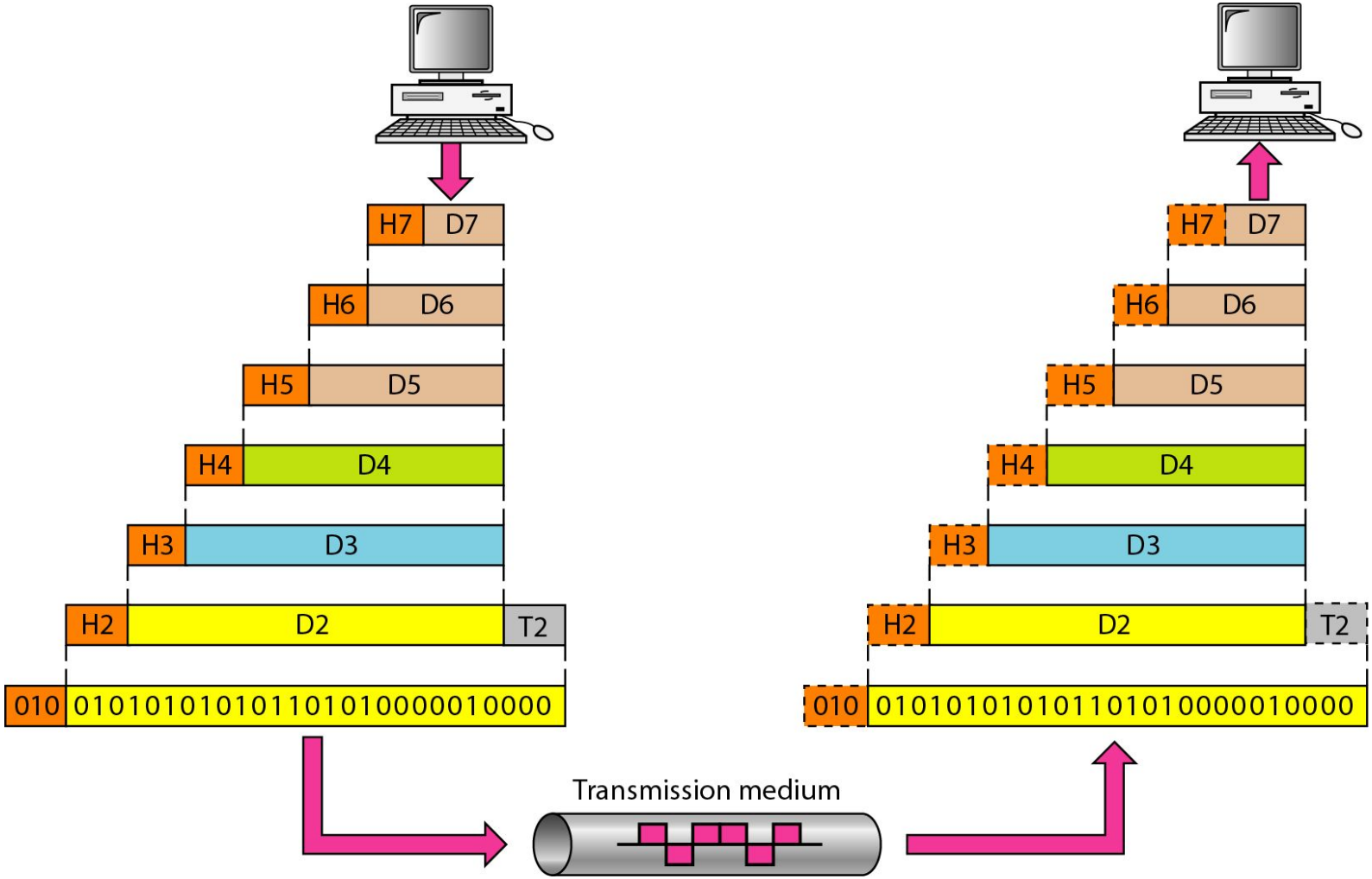


Figure 2.4 *An exchange using the OSI model*



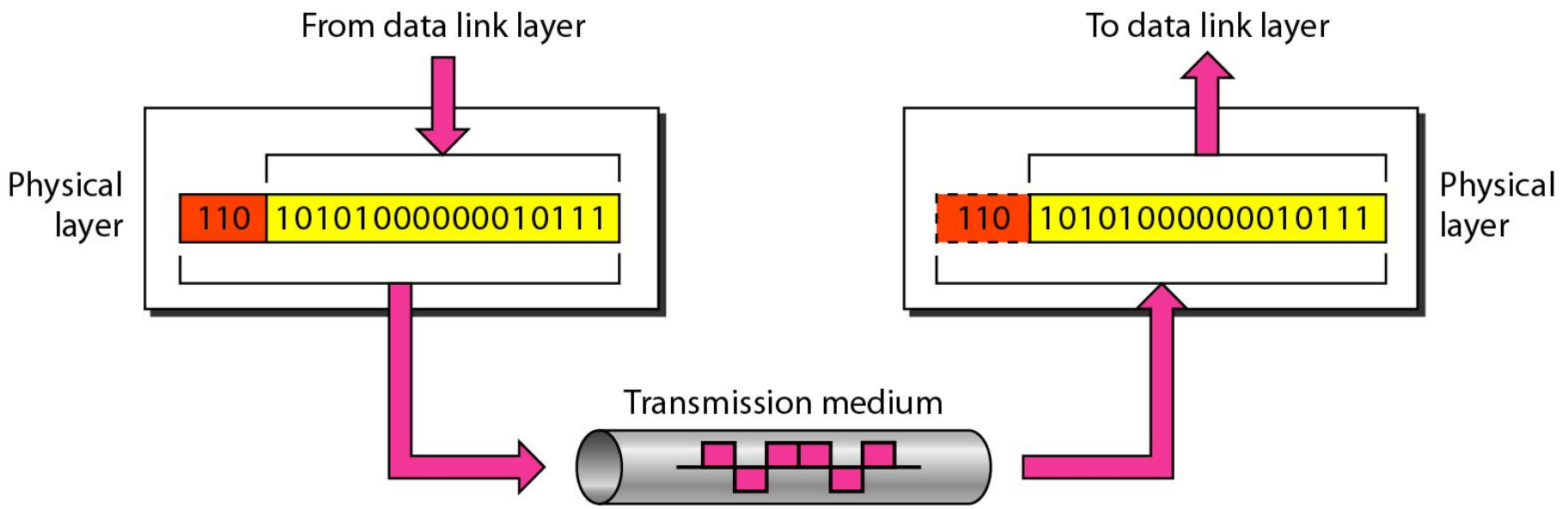
2-3 LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

Physical Layer
Data Link Layer
Network Layer
Transport Layer
Session Layer
Presentation Layer
Application Layer

Figure 2.5 *Physical layer*



The physical layer coordinates the functions required to carry a bit stream over a physical medium.

Functions of Physical layer:

Physical characteristics of interfaces and medium. The physical layer defines the characteristics of the interface between the devices and the transmission medium.

Representation of bits. To be transmitted, bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding (how 0s and 1s are changed to signals).

Data rate. The transmission rate-the number of bits sent each second-is also defined by the physical layer.

Synchronization of bits. The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level.

Line configuration. The physical layer is concerned with the connection of devices to the media.

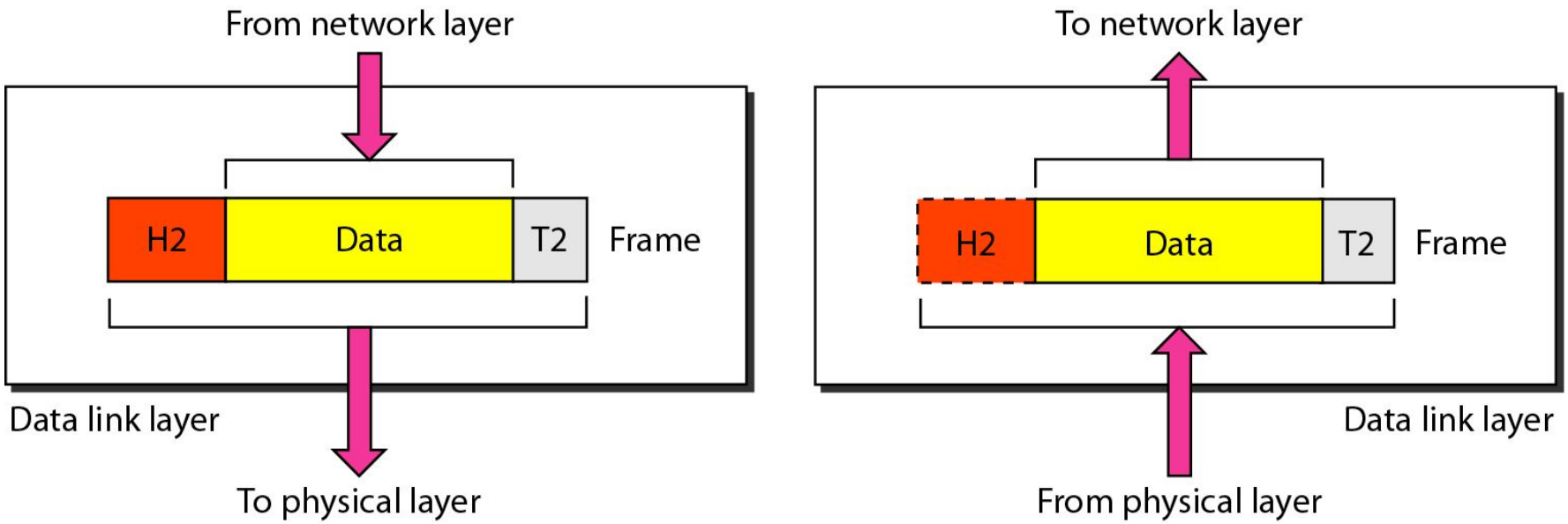
Physical topology. The physical topology defines how devices are connected to make a network.

Transmission mode. The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Figure 2.6 *Data link layer*



The data link layer is responsible for moving frames from one hop (node) to the next.

Other responsibilities of the data link layer include the following:

Framing. The data link layer breaks the message received from the network layer into streams of bits called as frames.

Physical addressing. If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.

Flow control. If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.

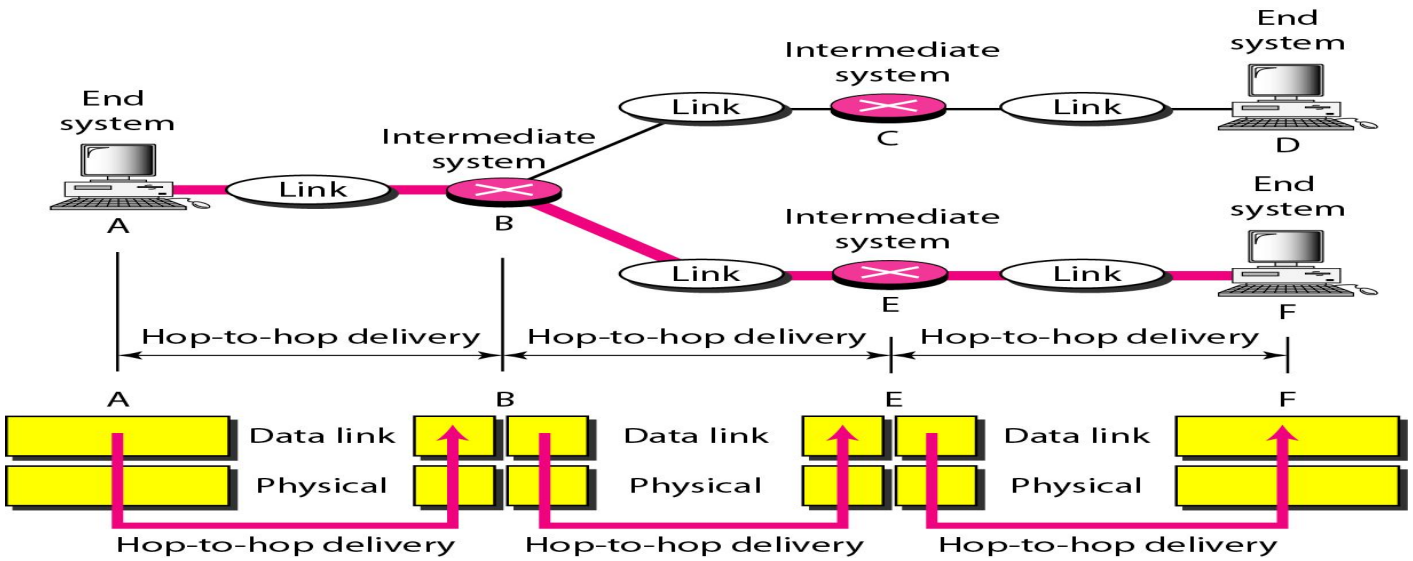
Error control. The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames.

Access control. When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

Note

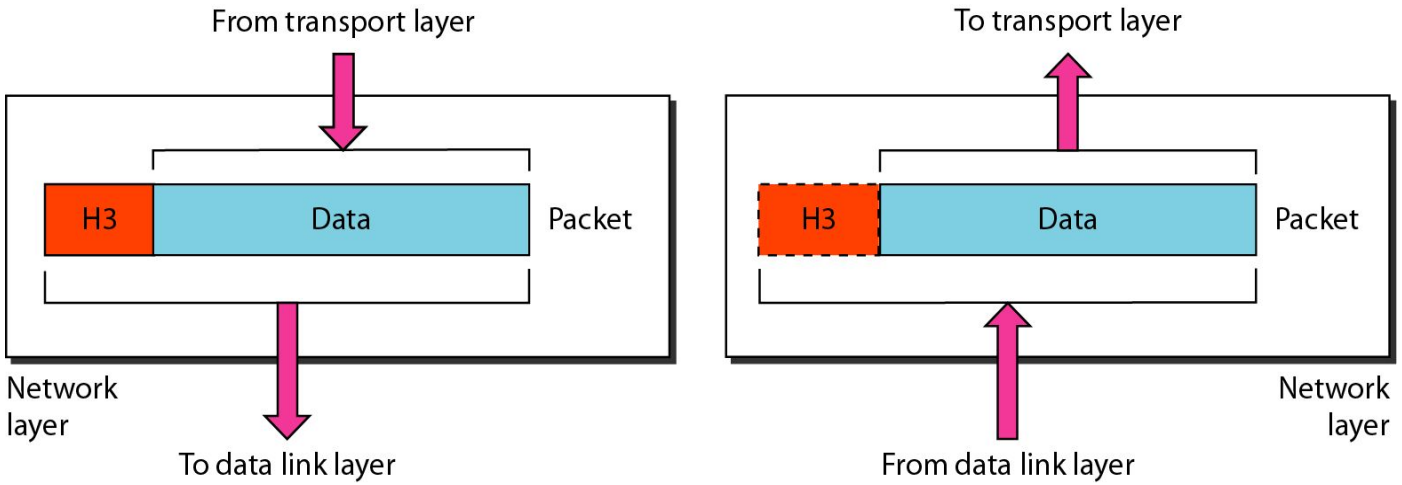
The data link layer is responsible for moving frames from one hop (node) to the next.

Figure 2.7 *Hop-to-hop delivery*



As the figure shows, communication at the data link layer occurs between two adjacent nodes. To send data from A to F, First, the data link layer at A sends a frame to the data link layer at B (a router). Second, the data link layer at B sends a new frame to the data link layer at E. Finally, the data link layer at E sends a new frame to the data link layer at F.

Figure 2.8 *Network layer*



The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.

Other responsibilities of the network layer include the following:

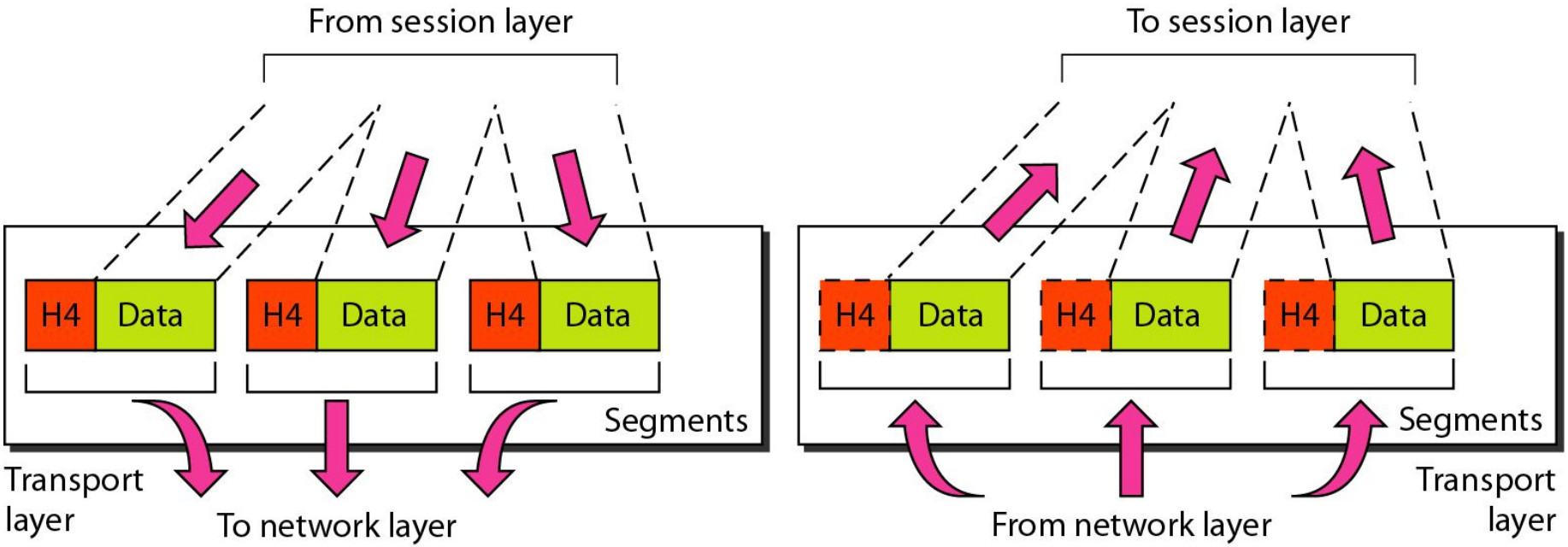
Logical addressing. The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Routing. When independent networks or links are connected to create *internetworks* (network of networks) or a large network, the connecting devices (called *routers* or *switches)* route or switch the packets to their final destination.

Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Figure 2.10 *Transport layer*



Other responsibilities of the transport layer include the following:

Service-point addressing. Source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a *service-point address* (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.

Segmentation and reassembly. A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

Connection control. The transport layer can be either connectionless or connection oriented.

Flow control. Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.

Error control. Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to-process rather than across a single link.

Note

The transport layer is responsible for the delivery of a message from one process to another.

Figure 2.11 *Reliable process-to-process delivery of a message*

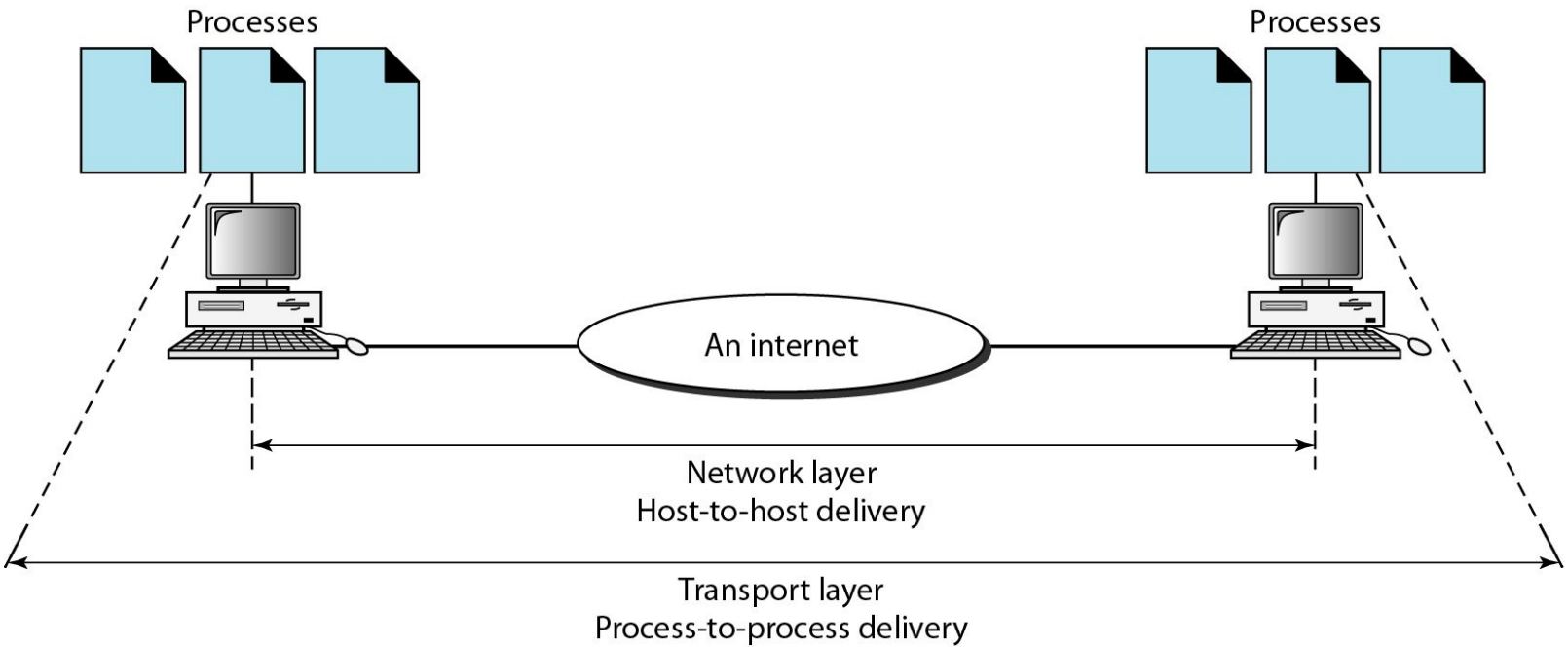
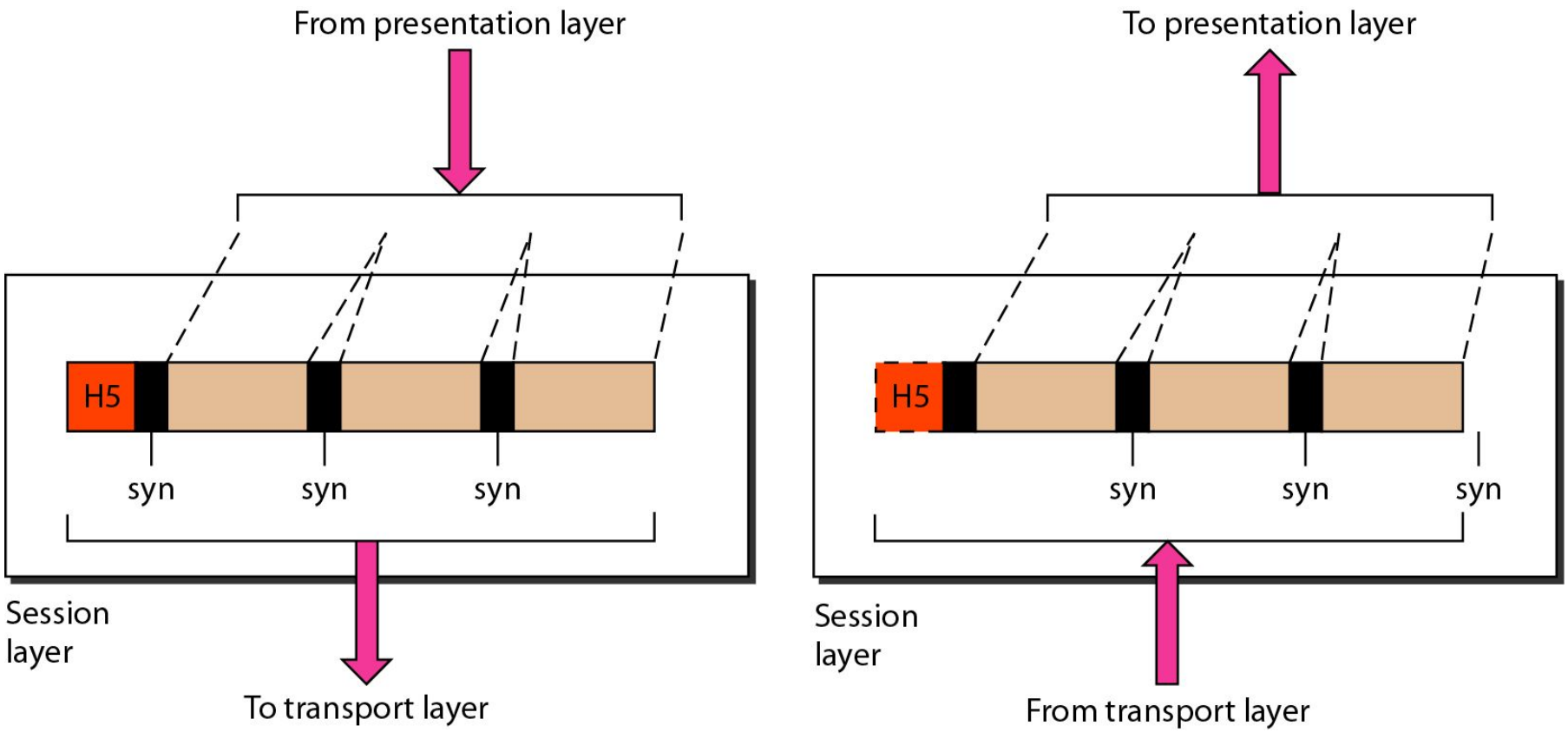


Figure 2.12 *Session layer*



The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes. The session layer is the network *dialog controller.*

It establishes, maintains, and synchronizes the interaction among communicating systems.

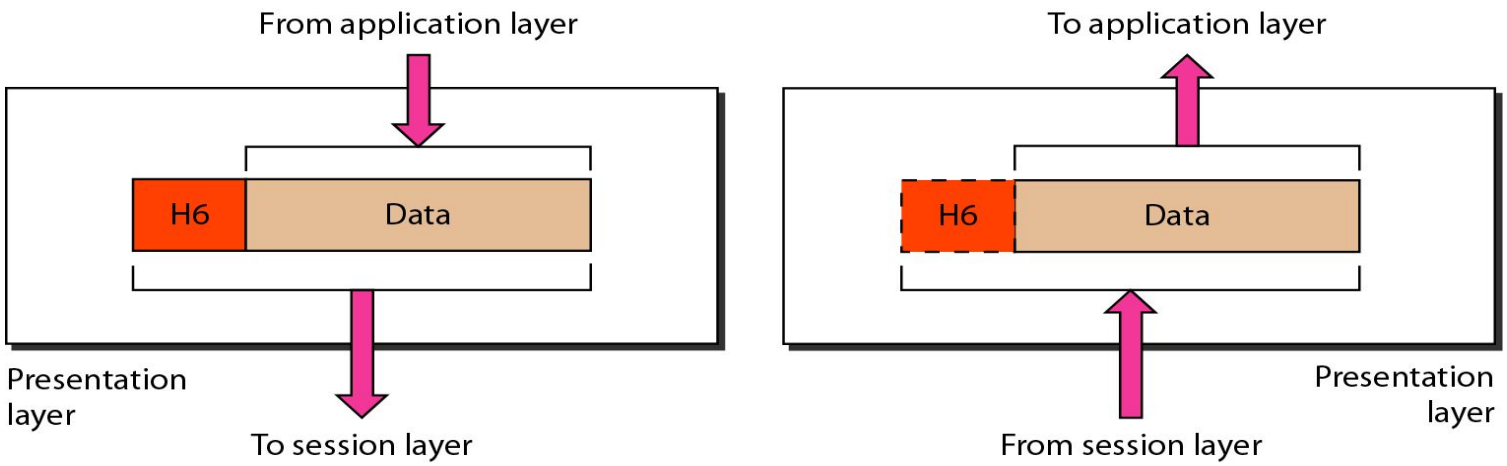
Specific responsibilities of the session layer include the following:

- Dialog control.
- Synchronization.

Note

The session layer is responsible for dialog control and synchronization.

Figure 2.13 *Presentation layer*



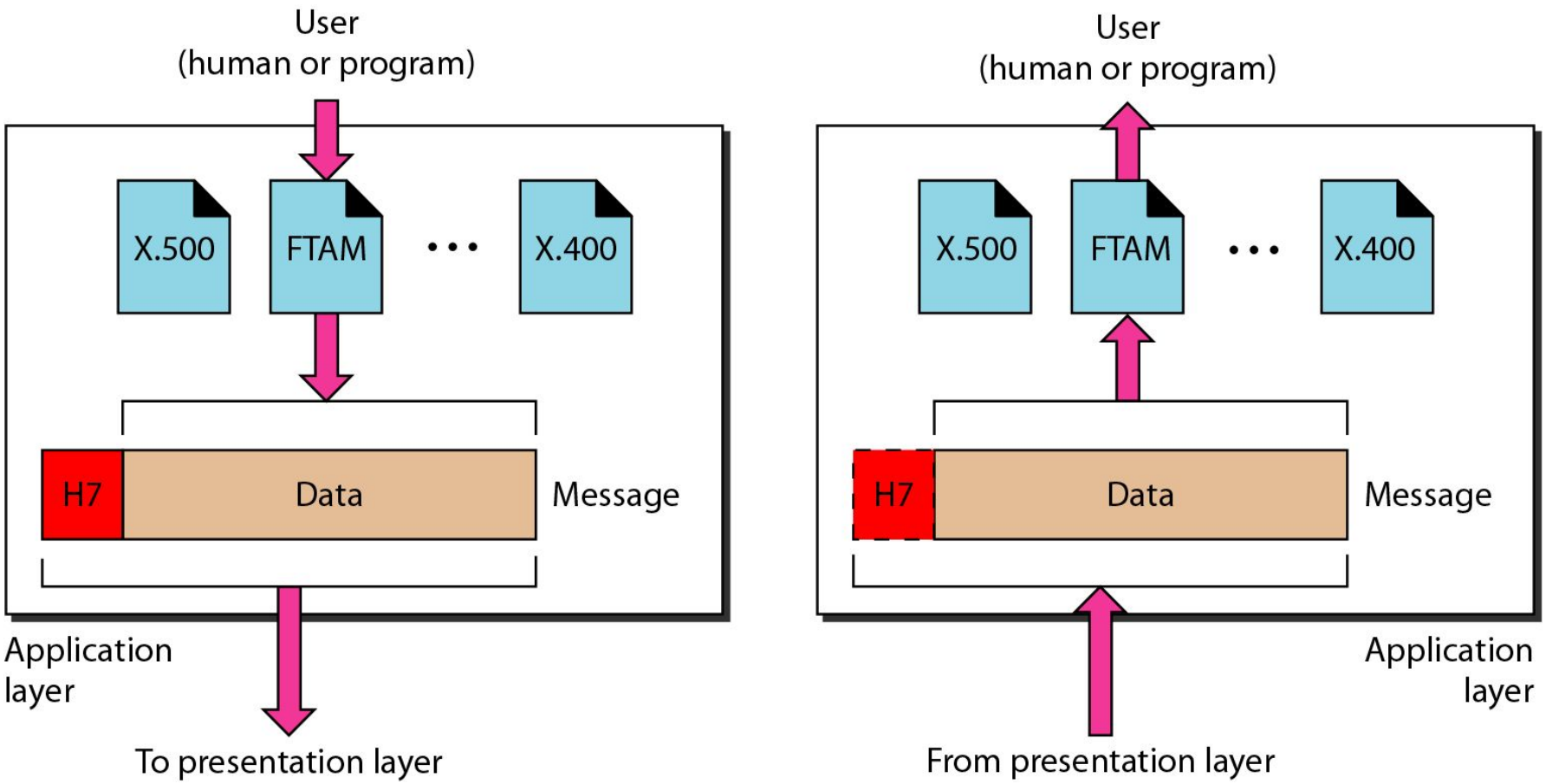
Functions:

1. Translation
2. Encryption
3. Compression

Note

The presentation layer is responsible for translation, compression, and encryption.

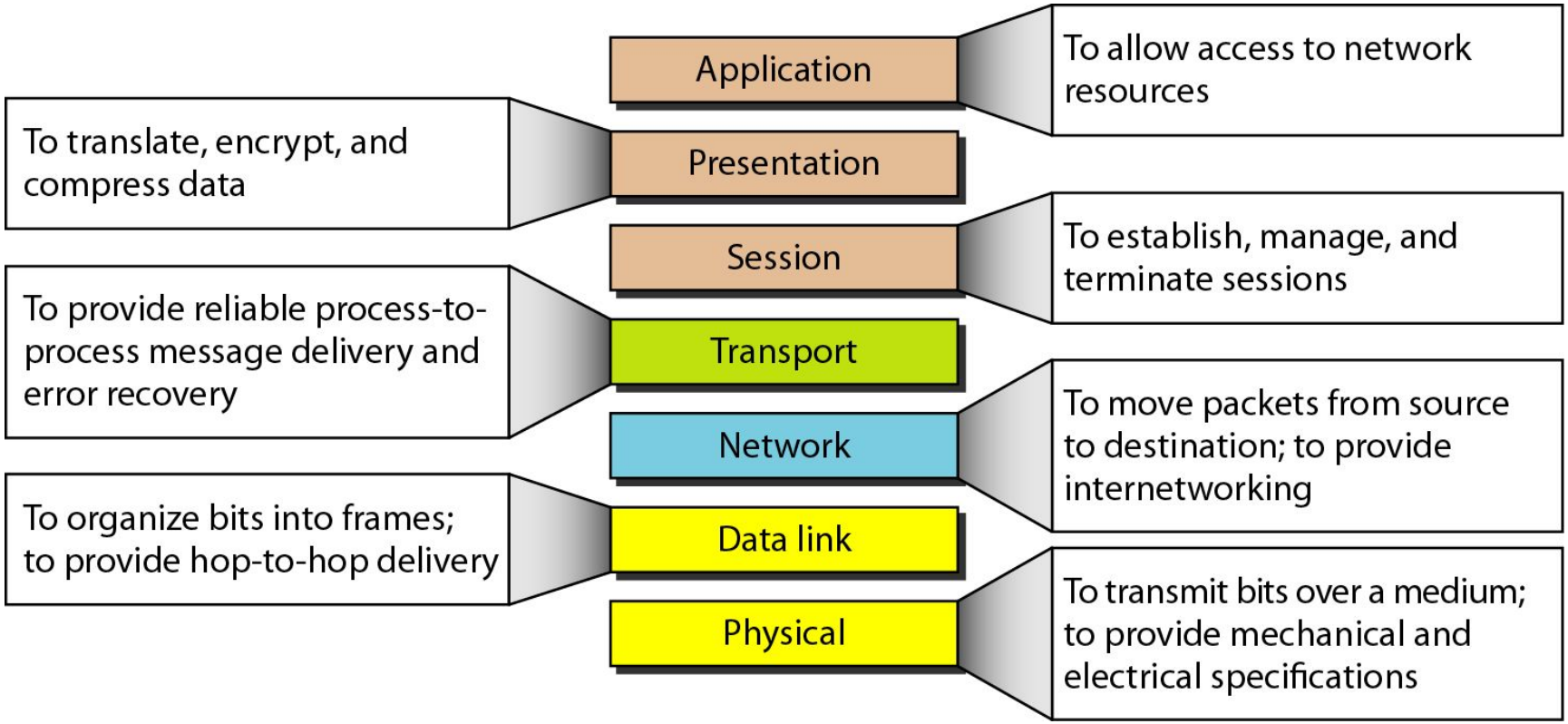
Figure 2.14 *Application layer*



Note

The application layer is responsible for providing services to the user.

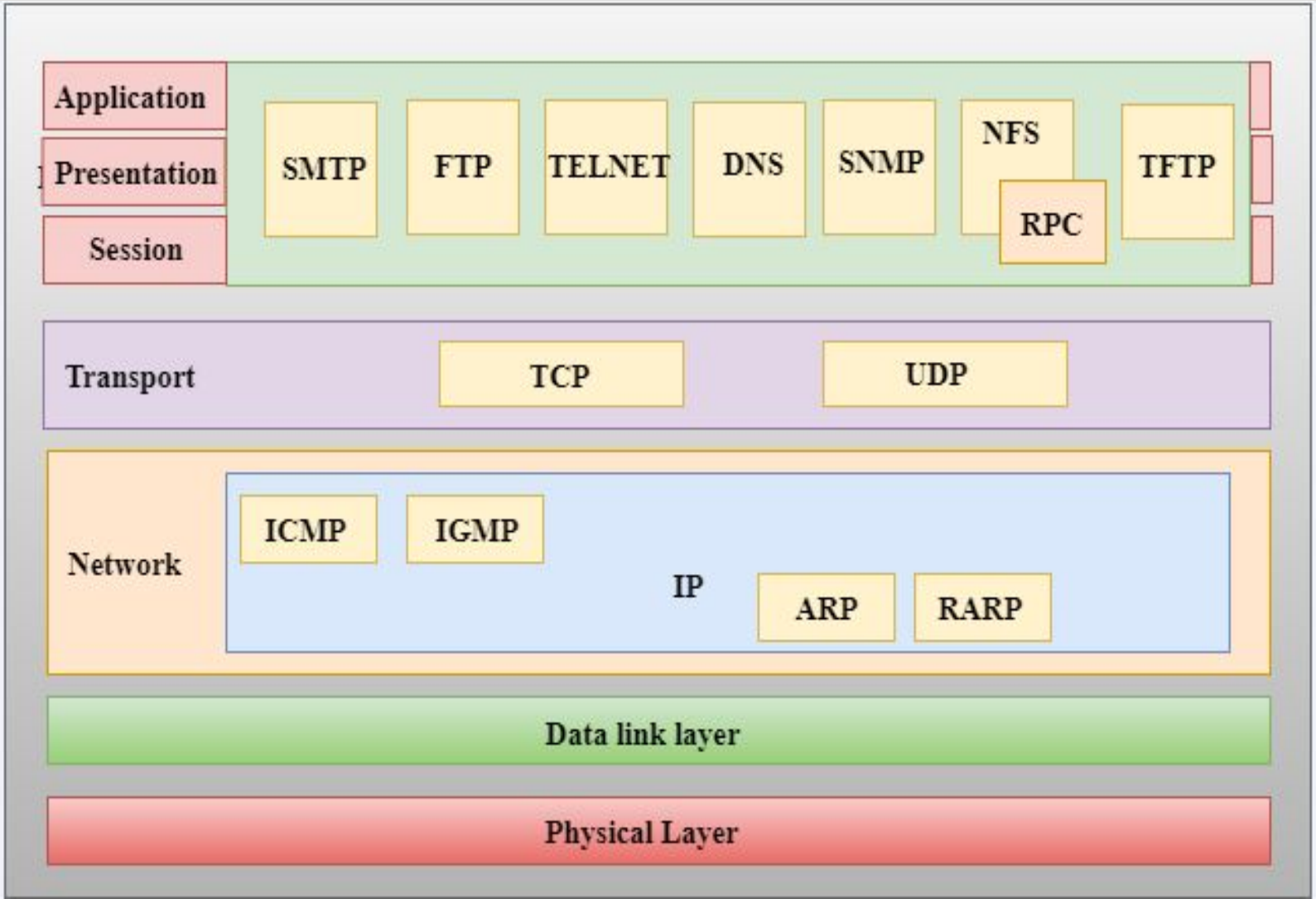
Figure 2.15 *Summary of layers*



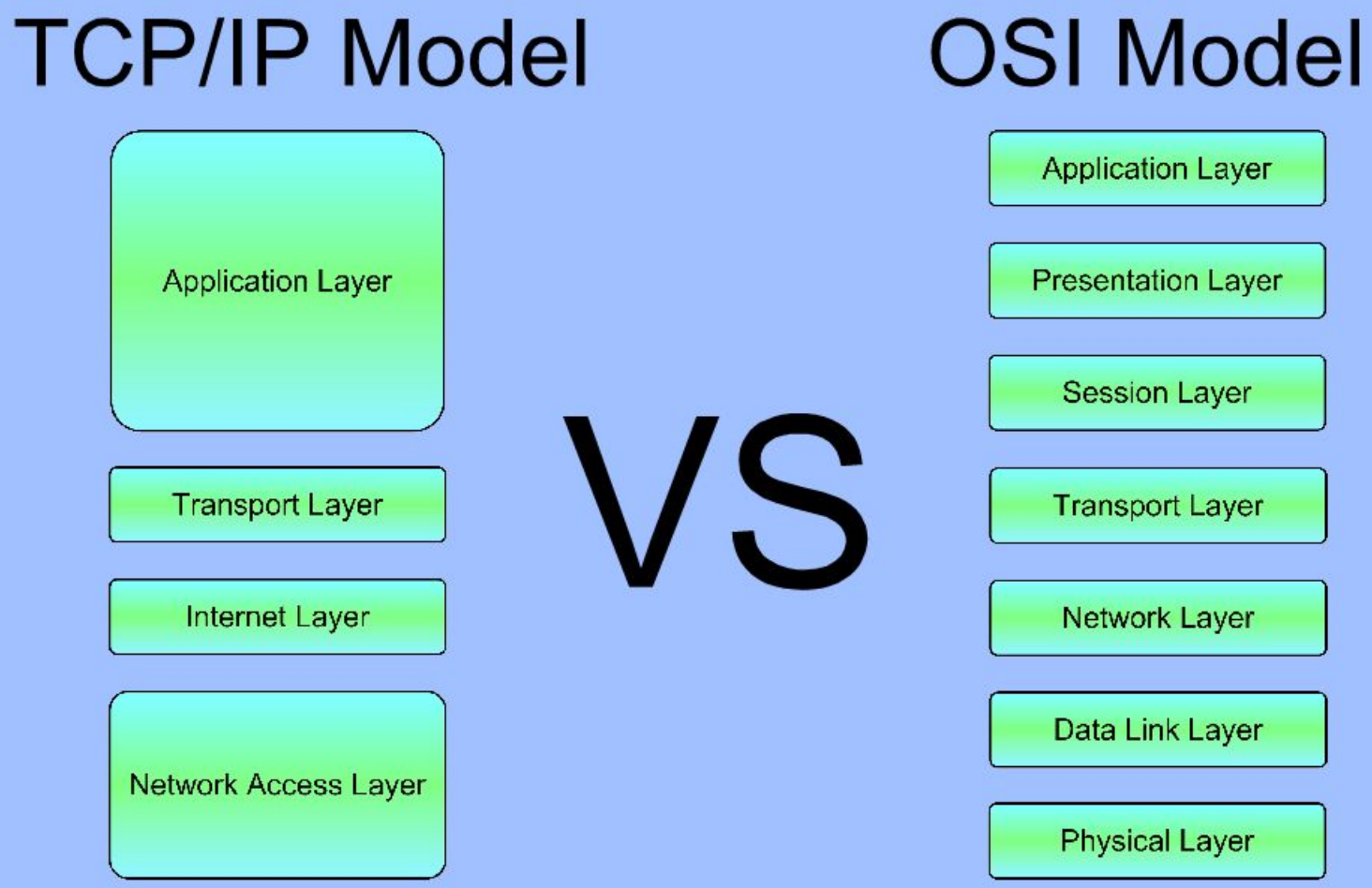
TCP/IP Protocol stack

- **The TCP/IP model was developed prior to the OSI model.**
- **The TCP/IP model is not exactly similar to the OSI model.**
- **The TCP/IP model consists of five layers: the application layer, transport layer, network layer, data link layer and physical layer.**
- **The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model and top three layers are represented in TCP/IP model by a single layer called the application layer.**
- **TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.**

TCP/IP Protocol stack



TCP/IP Protocol stack



Network Access Layer

Role
Handles actual transmission of data over physical media.

Responsibilities
Frame creation
MAC addressing
Error detection
Physical signaling

Technologies Used
Ethernet
Wi-Fi
Fiber networks
Cellular networks

Internet Layer

Role

Responsible for addressing and routing packets across networks.

Responsibilities

Logical addressing

Packet routing

Fragmentation

Path determination

Key Protocols

IP (Internet Protocol)

Provides logical addresses (IP addresses).

Routes packets from source to destination.

Versions: IPv4, IPv6

ICMP

Used for error reporting and diagnostics.

Example: ping command.

ARP

Maps IP addresses to MAC addresses.

Transport Layer

Role

Ensures communication between applications running on different hosts.

Responsibilities

Process-to-process delivery

Segmentation of data

Flow control

Error control

Port addressing

Reliability management

Key Protocols: TCP, UDP

Application Layer

Role
The Application Layer provides services directly to user applications. It combines the functions of the **Application, Presentation, and Session layers** of the OSI model.

Responsibilities
User interaction with network services
Data formatting
Session establishment
File transfer
Email communication

Common Protocols
HTTP/HTTPS – Web browsing
FTP – File transfer
SMTP – Sending emails
POP3– Receiving emails
DNS – Domain name resolution
Telnet– Remote login

Data Flow in TCP/IP Stack

When sending data:

Sender Side
Application creates data.
Transport layer segments data.
Internet layer adds IP address.
Network layer sends frames physically.

Receiver Side
Frame received.
IP layer checks destination.
Transport layer reassembles data.
Application receives message.